

Positivity of the leading log coefficient

Claude, with Daniel Murfet

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Abstract

Astra #12 rank 3. For equal starting exponents the leading log coefficient is $A_p = \frac{y_{00}}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2} e^{\beta s x_{00}} ds$ (unit 124). The Gaussian moment is strictly positive and monotone in x_{00} , so $y_{00} > 0$ implies $A_p > 0$ (`leading_log_coeff_pos`), with the quantitative bound $A_p \geq \frac{c}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2} e^{-\beta s M} ds$ when $y_{00} \geq c \geq 0$ and $\|x\|_\rho \leq M$ (`leading_log_coeff_ge`); both are restated on the coefficient-pair space. This is the nonvanishing of the denominator in the posterior ratio $Z_n[\varphi]/Z_n$ of `cor:empirical_expectation`. Zero `sorry/axiom`.

1 The things formalised

```
theorem gaussMoment_exp_integrableOn ( x ) (h : 0 < ) (h : 0 < ) :
  IntegrableOn (fun s => s^(-1) * log s ^ 0 * (exp (- *s^2) * exp (*s*x))) (Ioi 0)
theorem logMoment_exp_pos ( x ) (h) (h) : 0 < logMoment 0 (fun s => exp ( * s * x))
theorem logMoment_exp_mono ( x x') (h) (h) (hxx' : x x') :
  logMoment 0 (fun s => exp (*s*x)) logMoment 0 (fun s => exp (*s*x'))
theorem leading_log_coeff_pos ( ) (h h k k) (h hk hk) (p) (hp hp) (x y)
  (hy : 0 < y (0,0)) :
  0 < canonA h h k k (fun i j s => ampCoeff x y (i, j) s) p
theorem leading_log_coeff_ge ( M c) ... (hx : WSummable x) (hM : wnorm x M)
  (hc : 0 < c)
  (hy : c < y (0,0)) :
  c / (k * k) * logMoment p 0 (fun s => exp ( * s * (-M))) canonA ... p
theorem coeffA_leading_pos ... (a : CoeffPair) (hy : 0 < toY a (0,0)) :
  0 < coeffA h h k k p a
theorem coeffA_leading_ge ... (ha : ||a|| M) (hy : c < toY a (0,0)) :
  ... coeffA h h k k p a
```

2 Mathematics

The integrand $s^{p-1} e^{-\beta s^2} e^{\beta s x}$ is positive on $(0, \infty)$ and integrable (Gaussian moment, unit 126), so its support meets $(0, \infty)$ in a set of infinite Lebesgue measure and the integral is positive. Monotonicity in x is pointwise. The lower bound on the ball uses $|x_{00}| \leq \|x\|_\rho \leq M$. As Astra #12 noted, positivity of η on the open box does not force $y_{00} > 0$; the hypothesis is stated at the corner ($y_{00} > 0$, or $y_{00} \geq c$), which is what positivity of η on the closed box gives.

3 Paper-facing meaning

`cor:empirical_expectation` divides the expansion of $Z_n[\varphi]$ by that of $Z_n = Z_n[1]$; the leading term is $A_{i_0, j_0}(\varphi, \xi_n)/B_{k_0, p_0}(\xi_n)$ and the paper cites the grey book for $B_{k_0, p_0} > 0$. At chart level with

equal starting exponents the denominator's leading coefficient is $A_p(x, y_1)$ with y_1 the Taylor data of the prior times Jacobian, positive at the corner; the present unit supplies its positivity and a quantitative lower bound uniform on balls of the fluctuation data, which is what the first-order division (next units) needs.

4 Lean notes

- `setIntegral_pos_iff_support_of_nonneg_ae hnn hint : 0 < ∫ f s) ;` the a.e. nonnegativity is stated as `(0 : ℝ) ≤ f`, so after `ae_restrict_of_forall_mem a` `show (0 : ℝ) ≤ ∫ f s` is needed before `positivity`; `Real.volume_Ioi` gives ∞ and `ENNReal.zero_lt_top` closes.
- `positivity` proves $0 < \int f s$ goals for products of positive factors once $0 < f s$ is in context.
- The file compiled on the first attempt.